The Job Sequencing and Tool Switching Problem

Part IV: A Cluster-Level MTZ Formulation for the Non-Compact SSP

Why it is not exact (job domination), the safe formulations to use, and the LP-strength trade-off

Scope and status. This note proposes replacing the exponentially many subtour-elimination constraints of the non-compact SSP formulation (Part II) by a polynomial ($O(n^2)$) family of cluster-level Miller–Tucker–Zemlin (MTZ) constraints. Two facts must be stated up front, because they determine how the formulation may legitimately be used.

(1) The model is *not* exact (now settled). The aggregate MTZ constraint does not merely forbid subtours: it forbids any solution that traverses an ordered cluster pair twice and, through the job-indexed potentials u_j, any solution whose induced precedence digraph has a cycle (section 4.2). It thus optimises over a strict subset of schedules, so $\mathrm{OPT}_{\mathrm{MTZ}} \geq \mathrm{OPT}_{\mathrm{SSP}}$ — a heuristic upper bound. We previously left exactness (the "exchange argument") open; it is now **resolved negatively**: there is a $b = 3$ counterexample (theorem 4.4) on which $\mathrm{OPT}_{\mathrm{MTZ}} = 2 > 1 = \mathrm{OPT}_{\mathrm{SSP}}$, and the failure is generic (it is triggered by *job domination* $T_i \subseteq T_j$, which is ubiquitous). **Therefore this formulation must not be used as an exact model for column generation.** The safe exact alternatives — per-configuration MTZ and lazy SEC/PCTSP cuts — are given in section 7; *per-configuration MTZ exactness is now proved* (theorem 7.3). This revision also fixes two flaws found on verification: the coverage constraints (7)–(8) must sum over V^+, not V (else paths starting or ending inside a singleton cluster are wrongly cut off), and the raw ILP admits *parasitic cycles* that fake coverage (lemma 4.9), so even heuristic use requires path extraction. The aggregate Desrochers–Laporte lifting claimed earlier is invalid (remark 5.3).

(2) The motivation is a trade-off, not a free win. Removing row generation costs LP strength: MTZ-type subtour constraints give a relaxation no tighter than SEC, and aggregation to clusters weakens it further (remark 5.1). Whether the saved separation outweighs the looser bound inside branch-and-price is an *open computational question* (section 5); this note claims no net improvement.

Contents

Motivation: The Double-Generation Problem and the MTZ Trade-off

The non-compact configuration ILP of Part II (`02_noncompact_formulation.tex`) has two simultaneously exponential components:

- **Exponential columns**: $|V| = \binom{|T|}{b}$ configuration/arc variables, handled by **column generation** (a set-union knapsack pricing problem);
- **Exponential rows**: subtour-elimination constraints (SEC/GSEC), handled by **row generation** (separation).

Running both at once ("double generation") is awkward: a newly priced column can violate previously satisfied subtour cuts, forcing repeated separation passes and complicating the branch-and-price bookkeeping [Barnhart et al.(1998)Barnhart, Johnson, Nemhauser, Savelsbergh, and Vance, Lübbecke and Desrosiers(2005)].

The proposal. Replace the SECs by $O(n^2)$ cluster-level MTZ constraints [Miller et al.(1960)Miller, Tucker, and Row generation then disappears; only column generation remains. The constraint-side comparison is:

	SEC-based (Part II)	MTZ-based (this note)
Variables (x_{ij})	$\lvert V\rvert^2$ (exponential)	$\lvert V\rvert^2$ (exponential)
Subtour constraints	$\sim 2^{\lvert V\rvert}$	$n(n-1)$ (polynomial)
Coverage constraints	n	$2n$
Column generation?	Yes	Yes
Row generation?	**Yes**	**No**
LP relaxation strength	**stronger**	weaker (remark 5.1)
Captures every SSP optimum?	Yes	*Conditional* (theorem 4.11)

What is gained and what is paid. The last two rows are the crux and were glossed over in earlier drafts. Removing row generation is genuine. But (i) MTZ subtour constraints give a weaker LP relaxation than SEC, and the cluster aggregation weakens it further (remark 5.1); and (ii) the aggregate constraint additionally *restricts* the integer feasible set, so the model can miss SSP optima unless the exchange argument holds (section 4.2). The proposal thus trades a tight, exact SEC model (much separation) for a cheap, conditionally exact MTZ model (no separation, weaker bound). Which wins inside branch-and-price is empirical; we flag it in section 5. The Desrochers–Laporte lifting (remark 5.3) recovers part of the lost LP strength at the same constraint count.

Notation

- $J = \{1, \dots, n\}$: jobs; T: tools; b: magazine capacity.
- $T_j \subseteq T$, $|T_j| \le b$: mandatory tools of job j.
- V: all valid configurations — subsets $\mathcal{C} \subseteq T$ with $|\mathcal{C}| = b$. $|V| = \binom{|T|}{b}$, exponential in general.
- Configuration cluster of job j^*: $H_{j^*} = \{\mathcal{C} \in V : T_{j^*} \subseteq \mathcal{C}\}$, the configurations that can serve j^*. Clusters may overlap.
- Switching cost between $i, j \in V$: $d_{ij} = b - |i \cap j| = |i \setminus j|$.
- Dummy source/sink $s, t \notin V$ for the path formulation (zero-cost arcs to/from all of V).

The Cluster-Level GTSP-MTZ Formulation

Let $V^+ = V \cup \{s,t\}$ and $x_{ij} \in \{0,1\}$ for $(i,j) \in V^+ \times V^+$, $i \neq j$, with position variables $u_{j'}$ for each job $j' \in J$.

GTSP-MTZ formulation for the SSP (path version)

Objective:

$$\min \sum_{i \in V} \sum_{j \in V, j \neq i} d_{ij}\, x_{ij} \tag{1}$$

Path start/end:

$$\sum_{j \in V} x_{sj} = 1, \tag{2}$$

$$\sum_{i \in V} x_{it} = 1. \tag{3}$$

Degree (each config used at most once):

$$\sum_{j \in V^+ \setminus \{i\}} x_{ij} \leq 1 \quad \forall i \in V, \tag{4}$$

$$\sum_{i \in V^+ \setminus \{j\}} x_{ij} \leq 1 \quad \forall j \in V. \tag{5}$$

Flow conservation:

$$\sum_{j \in V^+ \setminus \{i\}} x_{ij} = \sum_{j \in V^+ \setminus \{i\}} x_{ji} \quad \forall i \in V. \tag{6}$$

Cluster coverage — exit and entry of each job cluster (sums over V^+):

$$\sum_{i \in H_{j^*}} \sum_{j \in V^+ \setminus H_{j^*}} x_{ij} \geq 1 \quad \forall j^* \in J, \tag{7}$$

$$\sum_{i \in V^+ \setminus H_{j^*}} \sum_{j \in H_{j^*}} x_{ij} \geq 1 \quad \forall j^* \in J. \tag{8}$$

Cluster-level MTZ subtour elimination:

$$u_{j'} - u_{j''} + n \sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij} \;\leq\; n-1 \qquad \forall j', j'' \in J,\ j' \neq j''. \tag{9}$$

Bounds and integrality:

$$1 \leq u_{j'} \leq n \ \forall j' \in J, \qquad x_{ij} \in \{0,1\}\ \forall (i,j) \in A := (\{s\} \times V) \cup \{(i,j) \in V \times V : i \neq j\} \cup (V \times \{t\}). \tag{10}$$

Mathematical Verification

Objective and coverage

The objective (1) sums $d_{ij} = b - |i \cap j|$ over selected arcs; since $|i| = |j| = b$, $|i \setminus j| = b - |i \cap j|$ equals both the removals and (by the uniform model) the insertions, so (1) counts tool switches.

Proposition 4.1 (Coverage $\Leftrightarrow$ job feasibility)**.** *In any feasible solution whose arc support is a single s–t path (in particular, any parasitic-free solution, lemma 4.9), every job j^* is served by some visited configuration $\mathcal{C} \in H_{j^*}$.*

Proof. The selected arcs form an s–t path through configurations. Constraint (7) forces an arc leaving H_{j^*} and (8) an arc entering it; on a connected path the existence of both implies the path passes through at least one node of H_{j^*}, which serves j^*. □

The exchange argument: the central open problem

The aggregate MTZ constraint (9) carries a hidden assumption, which is the heart of this note.

Why (9) *restricts* rather than relaxes. The sum $\sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij}$ counts *all* arcs from cluster $H_{j'}$ to cluster $H_{j''}$. If this equals $k \geq 2$, (9) forces $u_{j''} \geq u_{j'} + n(k-1) + 1$, which with $1 \leq u \leq n$ is infeasible for $k \geq 2$. Hence the formulation *forbids* every solution that traverses some ordered cluster pair more than once. Because it removes solutions (rather than adding fractional ones), it is a *restriction*: it can only *raise* the optimum, never lower it. It is exact only if an optimal SSP solution never needs such a repeated cluster pair.

Definition 4.2 (Cluster-simple path)**.** A configuration path $C_{(1)}, \ldots, C_{(m)}$ is *cluster-simple* if for every ordered pair (j', j''), $j' \neq j''$, $\sum_{l=1}^{m-1} \mathbf{1}[\, C_{(l)} \in H_{j'},\, C_{(l+1)} \in H_{j''}\,] \leq 1$.

Proposition 4.3 (Exact representability condition)**.** *A configuration path is representable by (9) (for some $u \in [1, n]$)* iff *(i) every ordered cluster pair is traversed at most once (definition 4.2), and (ii) the precedence digraph D — edge $j' \to j''$ whenever some arc runs from a config in $H_{j'}$ to one in $H_{j''}$ — is* acyclic.

Proof. Two arcs on a pair give sum ≥ 2 and infeasibility (above), forcing (i). Given (i), (9) reduces to $u_{j'} < u_{j''}$ per edge of D, solvable in $[1, n]$ iff D is acyclic (topological order). □

□

Theorem 4.4 (The cluster-MTZ formulation is *not* exact)**.** *There is a $b = 3$ instance with* $\mathrm{OPT}_{\mathrm{MTZ}} > \mathrm{OPT}_{\mathrm{SSP}}$*; the cluster-MTZ model excludes some optimal SSP solutions and is in general a strict over-estimate of* $\mathrm{OPT}_{\mathrm{SSP}}$*.*

Proof. Let $b = 3$, $T_1 = \{1, 4\}$, $T_2 = \{1, 3, 4\}$, $T_3 = \{1\}$, $T_4 = \{1, 2, 4\}$ ($U = \{1, 2, 3, 4\}$). The only config covering T_2 is $\{1, 3, 4\}$ and the only one covering T_4 is $\{1, 2, 4\}$, so every covering path uses both; they already cover all four jobs and $d(\{1, 3, 4\}, \{1, 2, 4\}) = 1$, hence $\mathrm{OPT}_{\mathrm{SSP}} = 1$ (one arc). On that arc both $T_3 = \{1\}$ and $T_1 = \{1, 4\}$ are covered at *both* endpoints (as $\{1\} \subseteq \{1, 4\}$ lie in both configs), so D contains $1 \to 3$ and $3 \to 1$: a 2-cycle. By proposition 4.3 no cost-1 path is representable; the cheapest representable covering path costs 2, so $\mathrm{OPT}_{\mathrm{MTZ}} = 2 > 1$. (Brute-force verified; 19 counterexamples in ≈ 7.9k instances, `plans-genai/_verification/`.) □ □

The failure is generic: job domination. Whenever $T_i \subseteq T_j$, every config covering j also covers i, so any arc with such a config on both ends puts $\{i, j\}$ on both sides and creates the 2-cycle $i \leftrightarrow j$ in D. Domination underlies the classical SSP dominance rules and is ubiquitous, so cluster-MTZ fails on a large fraction of realistic instances. **It is not a safe basis for an exact column-generation solver**; use the alternatives of section 7. The propositions below give the (restrictive) classes on which cluster-MTZ *does* stay exact.

Proposition 4.5 (Holds for pairwise-incompatible instances)**.** *If every pair of distinct jobs j', j'' has $|T_{j'} \cup T_{j''}| > b$ (so $H_{j'} \cap H_{j''} = \emptyset$), then every feasible configuration path — in particular every optimal one — is cluster-simple.*

Proof. With $H_{j'} \cap H_{j''} = \emptyset$, each configuration lies in at most one cluster, so each arc maps to a unique ordered cluster pair. The degree constraints (4)–(5) let each configuration appear once, so no ordered cluster pair receives two arcs. □

Proposition 4.6 (Holds for the 6-job ring)**.** *The* 6*-ring of Part V (`05_jgp_ssp_gap_analysis.tex`, Example 3.1) has a cluster-simple optimal solution.*

Proof. With $T_{j_k} = \{t_k, t_{k+1 \bmod 6}\}$, $b = 3$, the optimal sequence $j_1 \to \cdots \to j_6$ ($\mathrm{OPT}_{\mathrm{SSP}} = 3$, certified in Part V) uses the four configurations $\mathcal{C}_1 = \{t_1, t_2, t_3\}$, $\mathcal{C}_2 = \{t_1, t_3, t_4\}$, $\mathcal{C}_3 = \{t_1, t_4, t_5\}$, $\mathcal{C}_4 = \{t_1, t_5, t_6\}$, lying in clusters $\mathcal{C}_1 \in H_{j_1} \cap H_{j_2}$, $\mathcal{C}_2 \in H_{j_3}$, $\mathcal{C}_3 \in H_{j_4}$, $\mathcal{C}_4 \in H_{j_5} \cap H_{j_6}$. The three arcs hit exactly the ordered cluster pairs $(j_1, j_3), (j_2, j_3), (j_3, j_4), (j_4, j_5), (j_4, j_6)$, each once; all other 25 ordered pairs are unused. With $u_{j_1} = u_{j_2} = 1, u_{j_3} = 2, u_{j_4} = 3, u_{j_5} = u_{j_6} = 4$ one checks $u_{j'} - u_{j''} + 6 \cdot (\text{pair-sum}) \leq 5$ for all 30 ordered pairs. □

Remark 4.7 (Why these classes are safe but the general case is not)**.** Both classes avoid the domination 2-cycle of theorem 4.4. In the disjoint-tool case each configuration lies in a single cluster, so an arc induces exactly one ordered cluster pair and D is a simple path (acyclic). In the ring each optimal configuration serves at most the two ring-adjacent jobs and the optimal sequence visits clusters contiguously, so D is again acyclic. The general case fails precisely when a configuration covers a dominated pair $T_i \subseteq T_j$ on both ends of an arc (theorem 4.4); no hypothesis rules that out for arbitrary instances.

MTZ eliminates cluster-level subtours

Proposition 4.8 (No cluster-level subtours)**.** *If $\sum_{i \in H_{j'}} \sum_{j \in H_{j''}} x_{ij} \leq 1$ for all (j', j''), then no directed cycle visits configurations drawn only from clusters $\{H_{j'} : j' \in S\}$ for any $S \subsetneq J$.*

Proof. Suppose such a cycle exists. Summing (9) over its consecutive cluster pairs, the telescoping u-differences cancel and the arc term contributes $n|S|$ (one per pair), giving $n|S| \leq |S|(n-1)$, i.e. $n \leq n - 1$, a contradiction. □

Lemma 4.9 (Classification of surviving (parasitic) cycles)**.** *Let $Z = C_1 \to \cdots \to C_m \to C_1$ be a directed cycle in V carried by an integer solution. If every C_l lies in at least one cluster and the membership sets $\mathrm{cl}(C_l) = \{j \in J : C_l \in H_j\}$ are not all equal to one common singleton $\{j\}$, then (9) is infeasible for the solution. Hence exactly two families of cycles survive (9): (a) all configurations of Z have the identical singleton membership $\{j\}$; (b) Z contains a configuration covering no job.*

Proof. Choose for each node a cluster $a_l \in \mathrm{cl}(C_l)$; since the membership sets are not all the same singleton, either some $|\mathrm{cl}(C_l)| \geq 2$ or two nodes have different memberships, so the choice can be made non-constant around Z. Whenever $a_l \neq a_{l+1}$, the arc $C_l \to C_{l+1}$ contributes 1 to the pair sum of (a_l, a_{l+1}), and (9) forces $u_{a_l} \leq u_{a_{l+1}} - 1$; where $a_l = a_{l+1}$ no constraint is needed. Following Z once around, the inequalities chain into $u_{a_1} \leq u_{a_1} - r$ with $r \geq 1$ the number of change points ($r \geq 1$ since the cyclic sequence (a_l) is non-constant), a contradiction. □

Parasitic cycles: the ILP as written is not sound without post-processing. Family (a) of lemma 4.9 is harmless: every arc of such a cycle has both endpoints in H_j and in no other cluster, so it crosses no cluster boundary, contributes to no coverage row, and has strictly positive cost — it is absent from optima. Family (b) is *dangerous*: a cycle through an uncovered configuration can satisfy (7)–(8) for a job the s–t path never serves (*fake coverage*). Concretely, on the instance of theorem 4.4, the solution "path $s \to \{1,2,4\} \to t$ (cost 0) plus the disconnected 2-cycle $\{2,3,4\} \rightleftarrows \{1,3,4\}$ (cost 2)" satisfies every constraint of the model — the uncovered $\{2,3,4\}$ contributes no MTZ pair, and the cycle supplies entry and exit for H_2 — yet its path serves only jobs $1,3,4$ (verified constraint-by-constraint, `verify_mtz_doc.py`). Consequently $\mathrm{OPT}_{\mathrm{MTZ}} \geq \mathrm{OPT}_{\mathrm{SSP}}$ is *not* guaranteed for the raw ILP. Sound heuristic use requires *path extraction*: read off the s–t path, discard cycles, and reject (or repair) if some job is unserved. Restricting V to configurations covering at least one job eliminates family (b), but is *not* innocuous: uncovered configurations can be essential membership-resetting intermediates — on the theorem 4.4 instance every representable covering path passes through the uncovered $\{2,3,4\}$, so with covering-only V the model becomes infeasible there while $\mathrm{OPT}_{\mathrm{SSP}} = 1$. A full guarantee needs per-configuration MTZ (section 7, exactness proved in theorem 7.3).

Path structure and conditional correctness

Proposition 4.10 (Path structure)**.** *Constraints (2)–(6) and the degree bounds make the selected arcs the disjoint union of a single directed s–t path through a subset of V and (possibly) parasitic cycles of the two families of lemma 4.9; all other cycles are excluded by (9).*

Proof. One arc leaves s, one enters t; flow conservation and degree ≤ 1 at internal nodes give a disjoint union of one s–t path and cycles. lemma 4.9 eliminates every cycle except those of its families (a) and (b). □

Theorem 4.11 (Restriction; not exact in general)**.** *Restricted to* parasitic-free *solutions* *(lemma 4.9; enforced in practice by path extraction), the integer model* (1)–(10) *is a* restriction *of the SSP to* representable *schedules (cluster-simple with acyclic precedence digraph, proposition 4.3): every parasitic-free feasible integer solution is a feasible SSP schedule of equal cost, so the parasitic-free optimum* $\mathrm{OPT}_{\mathrm{MTZ}}$ *satisfies* $\mathrm{OPT}_{\mathrm{MTZ}} \geq \mathrm{OPT}_{\mathrm{SSP}}$ *and is the cheapest representable schedule. Equality* $\mathrm{OPT}_{\mathrm{MTZ}} = \mathrm{OPT}_{\mathrm{SSP}}$ *holds* iff *some optimal SSP schedule is representable; this* ***fails in general*** *(theorem 4.4) but holds for the classes of propositions 4.5 and 4.6.*

Proof. Every parasitic-free model solution is a schedule (hence $\mathrm{OPT}_{\mathrm{MTZ}} \geq \mathrm{OPT}_{\mathrm{SSP}}$*).* By propositions 4.1 and 4.10 a parasitic-free integer-feasible solution is a single s–t path that serves every job; processing each job at a covering configuration along the path gives an SSP schedule whose switch cost equals the objective (1). Thus a schedule of cost $\mathrm{OPT}_{\mathrm{MTZ}}$ exists, so $\mathrm{OPT}_{\mathrm{SSP}} \leq \mathrm{OPT}_{\mathrm{MTZ}}$. By proposition 4.8 and the over-constraining identity of section 4.2, the feasible solutions are exactly the cluster-simple schedules, so $\mathrm{OPT}_{\mathrm{MTZ}}$ is their minimum cost.

Conditional equality. If some optimal SSP schedule is *representable* (proposition 4.3), collapse its repeated configurations to $\mathcal{C}_{(1)}, \ldots, \mathcal{C}_{(k)}$, set $x_{s\mathcal{C}_{(1)}} = x_{\mathcal{C}_{(k)}t} = 1$, $x_{\mathcal{C}_{(l)}\mathcal{C}_{(l+1)}} = 1$, and take u from a topological order of the acyclic precedence digraph D. Coverage, degree and flow hold by construction; cluster-simplicity gives every ordered cluster-pair sum ≤ 1 and acyclicity makes the u-order consistent, so (9) holds, giving $\mathrm{OPT}_{\mathrm{MTZ}} \leq \mathrm{OPT}_{\mathrm{SSP}}$. When *no* optimal schedule is representable (theorem 4.4), $\mathrm{OPT}_{\mathrm{MTZ}} > \mathrm{OPT}_{\mathrm{SSP}}$ strictly. □

Reading of theorem 4.11. *After path extraction* (lemma 4.9) the model returns the cheapest *cluster-simple* schedule, which is a genuine SSP schedule — an *upper* bound on $\mathrm{OPT}_{\mathrm{SSP}}$. The raw ILP optimum may instead be a parasitic solution whose path misses jobs; always extract and check. When no optimum is cluster-simple, $\mathrm{OPT}_{\mathrm{MTZ}} > \mathrm{OPT}_{\mathrm{SSP}}$: the model *misses* the true optimum. A certified *lower* bound must therefore come from a genuine relaxation (the base degree/flow/coverage model, or the SEC formulation), *not* from the over-constrained MTZ model. Exactness *fails* in general (theorem 4.4).

Constraint Count, LP Strength, and Column Generation

Group	Description	Count
(2),(3)	Path start/end	2
(4),(5)	Config degree ≤ 1	$2\lvert V\rvert$
(6)	Flow conservation	$\lvert V\rvert$
(7),(8)	Cluster entry/exit coverage	$2n$
(9)	Cluster-level MTZ	$n(n-1)$
(10)	Position bounds	$2n$
Total	$(2n + n(n-1) + 2n + 2 = (n+1)(n+2))$	$\boldsymbol{3\lvert V\rvert + (n+1)(n+2)}$

The $|V|$-indexed constraints are handled by column generation; the job-indexed groups (coverage, MTZ, bounds) are $O(n^2)$ — polynomial, with no row generation. This is the intended structural gain over the $O(2^{|V|})$ SECs.

Remark 5.1 (LP strength: MTZ is no tighter than SEC)**.** For routing problems it is classical that compact MTZ subtour elimination yields an LP relaxation no tighter than — and generally strictly weaker than — the subtour-elimination (SEC/DFJ) relaxation, because MTZ enforces connectivity only through the potentials u [Applegate et al.(2006)Applegate, Bixby, Chvátal, and Cook, Desrochers and Laporte(1991)]. Two features make the present relaxation weaker still: the potentials are indexed by *jobs* (at most n values) rather than by configurations, and the per-arc terms are *aggregated* over $H_{j'} \times H_{j''}$, which coarsens the constraint. We therefore expect the root LP bound of this model to be looser than both the SEC model of Part II and the SSPMF bound $|U| - b$ of Part V; a precise polyhedral comparison for the cluster-aggregated system is left open.

> **Net effect (open, computational).** remark 5.1 says the root bound can only get looser, and theorem 4.11 says the integer model may miss optima. Against this, the model needs no separation and has $O(n^2)$ rows. Whether branch-and-price is faster overall — weaker bound and a possibly larger or truncated tree, but cheaper node LPs and no cut management — must be measured on the Catanzaro Tabela1C instances against the SEC/GSEC and SSPMF baselines. This note makes no such claim.

Remark 5.2 (Column-generation compatibility)**.** A newly priced arc $x_{i^*j^*}$ (with $i^* \in H_{j'}$, $j^* \in H_{j''}$) simply joins the existing MTZ constraint (9) for the pair (j', j'') with coefficient n; no new constraint is created. This is the standard behaviour of column generation — new variables acquire coefficients in existing rows [Lübbecke and Desrosiers(2005)] — and is exactly what avoids the SEC double-generation issue. The pricing problem seeks the arc of most negative reduced cost $\bar{d}_{ij} = d_{ij} - \mu_i - \lambda_j - \sum_{j^*: i \in H_{j^*}, j \notin H_{j^*}} \pi_{j^*}$.

Remark 5.3 (Desrochers–Laporte lifting recovers some strength)**.** For per-arc MTZ, Desrochers and Laporte [Desrochers and Laporte(1991)] strengthen $u_{j'} - u_{j''} + n\, x_{ij} \leq n - 1$ to $u_{j'} - u_{j''} + n\, x_{ij} + (n-2)x_{ji} \leq n-1$ at no extra constraint count. **Correction (this revision):** the natural aggregate analogue — replacing the arc terms by the forward and reverse cluster-pair sums — is *invalid*: it cuts off representable integer solutions. Example ($n = 3$, $b = 2$): $T_1 = \{1\}$, $T_2 = \{2\}$, $T_3 = \{3\}$, path $\{1, 4\} \to \{1, 2\} \to \{3, 5\}$. The forward pair sums force $u_1 \leq u_2 - 1 \leq u_3 - 2$, while the aggregate reverse term for the ordered pair $(3, 1)$ (the arc $\{1, 2\} \to \{3, 5\}$ runs from H_1 to H_3, so its reverse sum is 1) demands $u_3 - u_1 + (n - 2) \leq n - 1$, i.e. $u_3 - u_1 \leq 1$ — infeasible. The DL lifting is valid only per arc (because in a tour $x_{ji} = 1$ forces $u_i - u_j = 1$ exactly, an identity the cluster aggregation destroys); a per-arc MTZ here means $O(n^2|V|^2)$ constraints — exponential — and is not used. For the per-configuration model the per-arc DL lifting *is* valid (remark 7.4).

Structural Elements of the Path Formulation

(1) **Dummy source/sink** (2)–(3): the SSP is a path, not a Hamiltonian cycle, through the configuration graph; equivalently set $d_{ts} = 0$ and break the cycle at $(t \to s)$.
(2) **Flow conservation** (6): every entered internal configuration is exited, so the path cannot stop at an interior node.
(3) **Degree bounds** (4)–(5): simple path, no multigraph [Applegate et al.(2006)Applegate, Bixby, Chvátal, a
(4) **Cluster-level MTZ** (9): $O(n^2)$ subtour elimination in place of SECs, but *not exact* (theorem 4.4); LP-weaker than SEC (remark 5.1) and integer-restricting. Use section 7 for an exact solver.

> **Remark 6.1** (Why n is the right MTZ coefficient)**.** The variables $u_{j'}$ order *cluster visits*; there are $\leq n$ clusters, so $u_{j'} \in [1, n]$ and the coefficient n is tight. Using $|V|$ would only

weaken the relaxation.

What to Use Instead (for Exact Column Generation)

Since the cluster-MTZ model is not exact (theorem 4.4), an exact branch-and-price solver should use one of the following; both keep the configuration column generation intact and only change how subtours are eliminated.

(a) **Per-configuration MTZ (exact).** Give each configuration node $i \in V$ a potential $v_i \in [1, |V|]$ and impose, for generated arcs,

$$v_i - v_j + |V|\, x_{ij} \leq |V| - 1 \qquad \forall\, i, j \in V,\ i \neq j.$$

These are the *standard* MTZ subtour constraints on the configuration graph: they eliminate *all* subtours (lemma 7.1) — including the parasitic families of lemma 4.9 — and exclude no optimum; there is no domination pathology, because the potentials are per-configuration, not per-job, so the 2-cycle of theorem 4.4 cannot arise. Exactness is *proved* in section 7.1 (theorem 7.3). In column generation a new column i adds one variable v_i and arc rows over generated nodes only: $O(|V_{\text{gen}}|^2)$ rows — more than the $O(n^2)$ cluster form, but *correct*. (Use the sharper big-M of remark 7.4.) *Update (2026-06-10): for column generation this recommendation is superseded by the position-indexed master of Part X (`10_position_formulations.tex`) — proved exact with a* fixed $O(n|T|)$ *row set; per-configuration MTZ remains the choice when V is enumerated explicitly.*

(b) **Lazy SEC/PCTSP cuts (exact).** Keep the Part II degree/flow/coverage model and separate subtour cuts on integer incumbents (the standard double-generation loop); fully worked out in Part II.

The cluster-aggregated MTZ (9) remains usable as a fast *primal heuristic* (every feasible solution is a genuine schedule) or warm start — just not as the exact master. The job-indexed MTZ of the BBC master (Part VIII) is on *jobs*, not configurations, and is the standard exact ATSP MTZ, hence unaffected.

Exactness proof for the per-configuration MTZ model

On the arc set A of (10), the *per-configuration MTZ model* consists of (1)–(8) (coverage over V^+, as corrected) together with continuous potentials $v_i \in [1, |V|]$ for $i \in V$ and

$$v_i - v_j + |V|\, x_{ij} \;\leq\; |V| - 1 \qquad \forall i, j \in V,\ i \neq j, \tag{11}$$

in place of (9) and the u-variables. Write OPT_{cfg} for its optimum. Throughout, $d_{ij} = |i \setminus j|$ satisfies the triangle inequality $d_{ik} \leq d_{ij} + d_{jk}$ (since $i \setminus k \subseteq (i \setminus j) \cup (j \setminus k)$ and $|i| = |j| = |k| = b$).

Lemma 7.1 (All cycles eliminated)**.** *No solution of (11) with $v \in [1, |V|]^V$ carries a directed cycle within V.*

Proof. Summing (11) over the m arcs of a cycle, the potentials telescope to zero and each arc contributes $|V|$, giving $m|V| \leq m(|V| - 1)$, a contradiction. (Unlike (9), this applies to *every* cycle: the potentials live on configurations, so neither uncovered configurations nor shared cluster memberships create exempt arcs.) □

Lemma 7.2 (Normal form of optimal schedules)**.** *Every feasible SSP instance admits an optimal schedule whose magazine trajectory is a sequence of configurations $C_{(1)}, \ldots, C_{(m)}$ that are* pairwise distinct*, with $m \leq n$, each $C_{(l)}$ serving at least one job, and total switch cost $\sum_{l=1}^{m-1} d_{C_{(l)}C_{(l+1)}}$ (initial load $C_{(1)}$ free, dummy-depot convention).*

Proof. Take any optimal schedule. W.l.o.g. the magazine is full at every instant (pad arbitrarily; standard assumption (2) of Crama et al.), so the trajectory is a sequence of n configurations, one per job position, with switch cost $\sum |M_{l+1} \setminus M_l|$. (i) *Collapse* consecutive equal configurations; the collapsed terms contribute 0, so the cost telescopes unchanged and consecutive configurations are now distinct. (ii) *De-duplicate*: if $C_{(p)} = C_{(q)}$ with $p < q$, reassign the jobs served at visit q to visit p (the configuration is identical, so their tools remain loaded) and delete node q, joining its neighbours $X \to C_{(q)} \to Y$ into $X \to Y$; by the triangle inequality $d_{XY} \leq d_{XC_{(q)}} + d_{C_{(q)}Y}$ the cost does not increase. If this creates an adjacent equal pair ($X = Y$), collapse again. Each step strictly shortens the sequence, so the process terminates with pairwise distinct configurations. (iii) A configuration serving no job is deleted by the same shortcut without reassignment. Finally $m \leq n$: after (i)–(iii) every remaining configuration serves at least one job and every job is served exactly once. □

Theorem 7.3 (Per-configuration MTZ is exact)**.** $\mathrm{OPT}_{\mathrm{cfg}} = \mathrm{OPT}_{\mathrm{SSP}}$.

Proof. ($\mathrm{OPT}_{\mathrm{SSP}} \leq \mathrm{OPT}_{\mathrm{cfg}}$, soundness.) Let x, v be any integer-feasible solution. By (2)–(6) the support of x decomposes into one s–t path and directed cycles in V; lemma 7.1 excludes all cycles, so the support is a single simple path $s \to C_{(1)} \to \cdots \to C_{(k)} \to t$. By (8) (sums over V^+) every cluster H_{j^*} receives an arc, whose head is a node of the path, so every job is covered by a visited configuration. Assign each job to a covering visited configuration and order the jobs by the position of their assigned configuration (ties arbitrarily): this is an SSP schedule whose magazine trajectory is the path, with switch cost $\sum_l d_{C_{(l)}C_{(l+1)}}$ — the objective value (1) — after the free initial load $C_{(1)}$ (dummy-depot convention). Hence every feasible solution maps to a schedule of equal cost, and $\mathrm{OPT}_{\mathrm{SSP}} \leq \mathrm{OPT}_{\mathrm{cfg}}$.

($\mathrm{OPT}_{\mathrm{cfg}} \leq \mathrm{OPT}_{\mathrm{SSP}}$, completeness.) Take an optimal schedule in the normal form of lemma 7.2, with trajectory $C_{(1)}, \ldots, C_{(m)}$. Set $x_{sC_{(1)}} = x_{C_{(m)}t} = 1$ and $x_{C_{(l)}C_{(l+1)}} = 1$ for $1 \leq l \leq m-1$, all other $x = 0$; set $v_{C_{(l)}} = l$ and $v_i = |V|$ for every unused $i \in V$. Source/sink, degree, and flow conservation hold because the $C_{(l)}$ are pairwise distinct. (11): on a path arc, $v_i - v_j + |V| = (l) - (l+1) + |V| = |V| - 1$; on a non-arc pair, $v_i - v_j \leq |V| - 1$ since $v \in [1, |V|]$. Coverage: job j^* is served at some visited $C \in H_{j^*}$; let $l_{\min}$ (resp. $l_{\max}$) be the first (resp. last) position with $C_{(l)} \in H_{j^*}$. The predecessor of $C_{(l_{\min})}$ — $C_{(l_{\min}-1)}$ or s — lies in $V^+ \setminus H_{j^*}$ by minimality, giving (8); the successor of $C_{(l_{\max})}$ — $C_{(l_{\max}+1)}$ or t — gives (7). (This is exactly where the V^+ correction is needed: with sums over V only, a schedule whose first or last configuration is the unique member of some cluster would be wrongly excluded.) The objective equals $\sum_l d_{C_{(l)}C_{(l+1)}} = \mathrm{OPT}_{\mathrm{SSP}}$. Hence $\mathrm{OPT}_{\mathrm{cfg}} \leq \mathrm{OPT}_{\mathrm{SSP}}$. □

Remark 7.4 (Sharper big-M; valid per-arc DL lifting)**.** (i) By lemma 7.2 some optimal schedule uses $m \leq n$ distinct configurations, so (11) remains exact with $|V|$ replaced by $N := \min(n, |V|)$ and $v_i \in [1, N]$: soundness is unaffected (lemma 7.1 holds verbatim with N), and the completeness construction satisfies $v \le m \le N$. Paths longer than N configurations are cut off, but an optimum survives. Since typically $n \ll |V| = \binom{|T|}{b}$, this is a much tighter big-M for the LP relaxation. (ii) The per-arc Desrochers–Laporte lifting $v_i - v_j + N\,x_{ij} + (N-2)\,x_{ji} \le N - 1$ is *valid* here: for the path solution of theorem 7.3, $x_{ji} = 1$ implies $v_i - v_j = 1$, making the constraint tight, and all other cases are slack — contrast with the *invalid* cluster-aggregated lifting (remark 5.3).

Summary

- **A restriction, *not* exact.** The model optimises over representable schedules (cluster-simple with acyclic precedence digraph, proposition 4.3), so $\mathrm{OPT}_{\mathrm{MTZ}} \ge \mathrm{OPT}_{\mathrm{SSP}}$ always but $\mathrm{OPT}_{\mathrm{MTZ}} > \mathrm{OPT}_{\mathrm{SSP}}$ is possible: there is a $b = 3$ counterexample with $\mathrm{OPT}_{\mathrm{MTZ}} = 2 > 1 = \mathrm{OPT}_{\mathrm{SSP}}$ (theorem 4.4), and the failure is generic (job domination $T_i \subseteq T_j$). Exactness holds only on the restrictive classes propositions 4.5 and 4.6. Moreover the raw ILP admits *parasitic cycles* that fake coverage (lemma 4.9): heuristic use requires path extraction. **For an exact CG solver, use the per-configuration MTZ (now proved exact, theorem 7.3) or lazy SEC of section 7**, not the cluster form.
- **Polynomial subtour constraints, no row generation.** $O(n^2)$ MTZ constraints [Miller et al.(1960)Mil replace $\sim 2^{|V|}$ SECs; new columns never trigger separation (remark 5.2).
- **At an LP-strength cost.** The MTZ relaxation is no tighter than SEC, and the cluster aggregation weakens it further (remark 5.1); Desrochers–Laporte lifting (remark 5.3) recovers part of it. The net branch-and-price effect is an open computational question (section 5).
- **Still column-generated.** $|V| = \binom{|T|}{b}$ remains exponential; the model is meant for branch-and-price [Barnhart et al.(1998)Barnhart, Johnson, Nemhauser, Savelsbergh, and Vance, Lübbecke and I

Where This Fits: Solution Architecture

Formulation	Column gen.?	Row gen.?	Role
Compact ILPs (Part I)	No	No	exact, weak–medium LP
Non-compact + SEC (Part II)	Yes	**Yes**	exact, strong LP
MTZ (this note)	**Yes**	**No**	restriction; heuristic UB
Branch-and-Benders	No	Yes (lazy)	exact, polynomial subproblem

The MTZ route removes row generation, simplifying branch-and-price [Barnhart et al.(1998)Barnhart, Johns Lübbecke and Desrosiers(2005)], but inherits the classical MTZ-vs-SEC LP weakness [Applegate et al.(2006)Ap and additionally restricts the integer set. Until the exchange argument is settled it should be used as a heuristic that returns genuine schedules (upper bounds on $\mathrm{OPT}_{\mathrm{SSP}}$), with a certified lower bound taken from a true relaxation; it must be benchmarked against the SEC and SSPMF formulations before adoption as an exact solver.

References

[Applegate et al.(2006)Applegate, Bixby, Chvátal, and Cook] D. L. Applegate, R. E. Bixby, V. Chvátal, and W. J. Cook. *The Traveling Salesman Problem: A Computational Study*. Princeton University Press, 2006.

[Barnhart et al.(1998)Barnhart, Johnson, Nemhauser, Savelsbergh, and Vance] C. Barnhart, E. L. Johnson, G. L. Nemhauser, M. W. P. Savelsbergh, and P. H. Vance. Branch-and-price:

Column generation for solving huge integer programs. *Operations Research*, 46(3):316–329, 1998.

[Desrochers and Laporte(1991)] M. Desrochers and G. Laporte. Improvements and extensions to the Miller–Tucker–Zemlin subtour elimination constraints. *Operations Research Letters*, 10(1):27–36, 1991.

[Lübbecke and Desrosiers(2005)] M. E. Lübbecke and J. Desrosiers. Selected topics in column generation. *Operations Research*, 53(6):1007–1023, 2005.

[Miller et al.(1960)Miller, Tucker, and Zemlin] C. E. Miller, A. W. Tucker, and R. A. Zemlin. Integer programming formulation of traveling salesman problems. *Journal of the ACM*, 7(4): 326–329, 1960.